Lab Report: Comparing CNN architectures

1 Introduction

In this report we compare several models on standard benchmarks. We trained each model for 30 epochs with a batch size of 128. The learning rate was set to 0.01 with the Adam optimizer.

2 Results

Figure 1 shows the test accuracy during training.

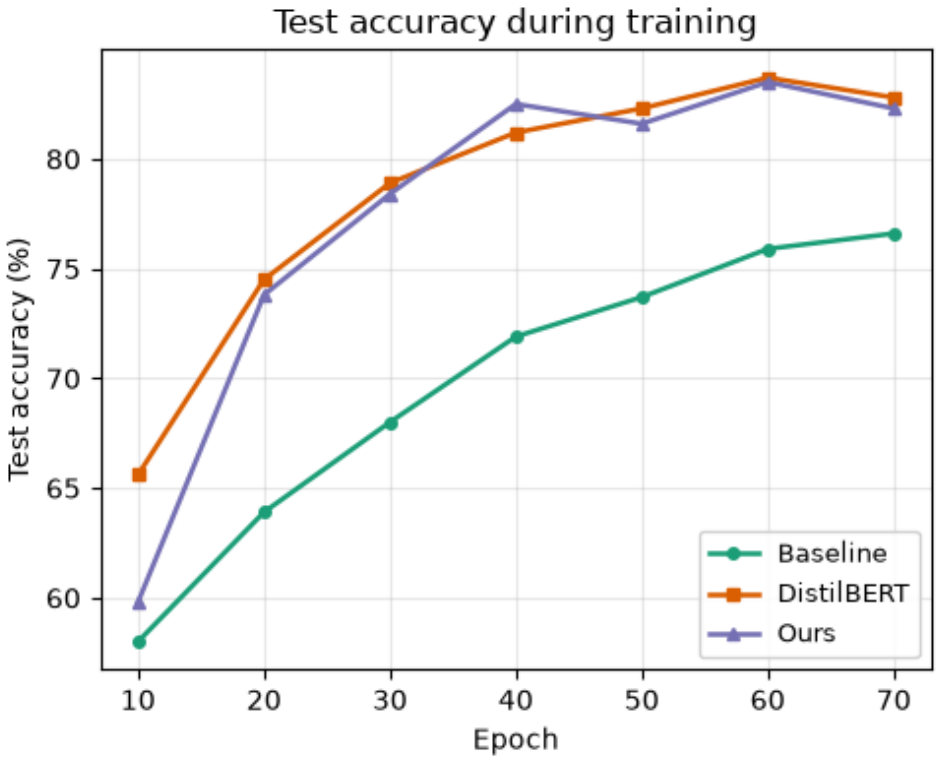


Figure 1: Test accuracy during training for the models in this experiment.

As shown in Figure 1, at epoch 60, Baseline achieves 83.7% test accuracy. DistilBERT outperforms Baseline by 11.2 percentage points at epoch 20 (Figure 1). As shown in Figure 1, at epoch 20, DistilBERT achieves 74.5% test accuracy.

Figure 2 shows the test accuracy by dataset.

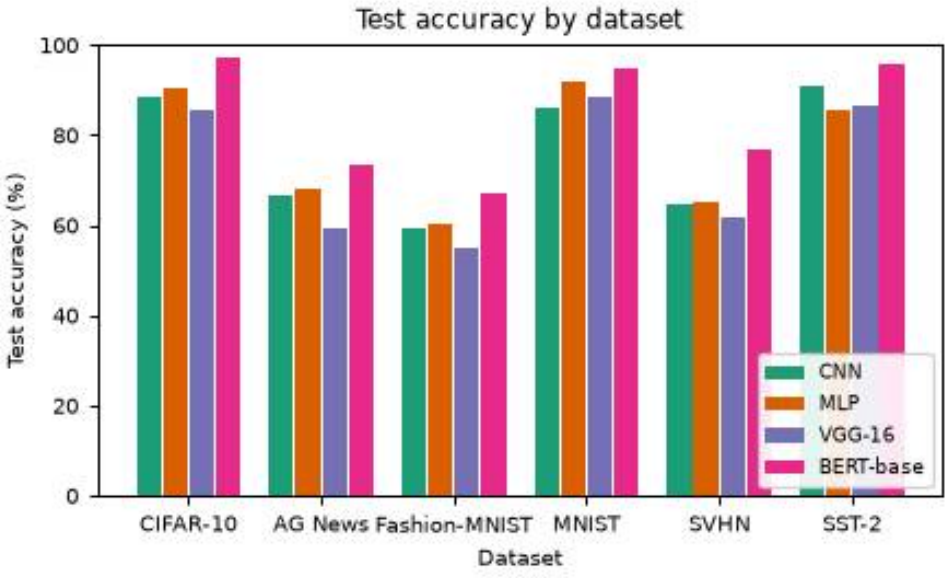


Figure 2: Test accuracy by dataset for the models in this experiment.

CNN reaches 100.8% test accuracy on SST-2. As shown in Figure 2, MLP outperforms VGG-16 by 5 percentage points on CIFAR-10.

Figure 3 shows the test accuracy during training.

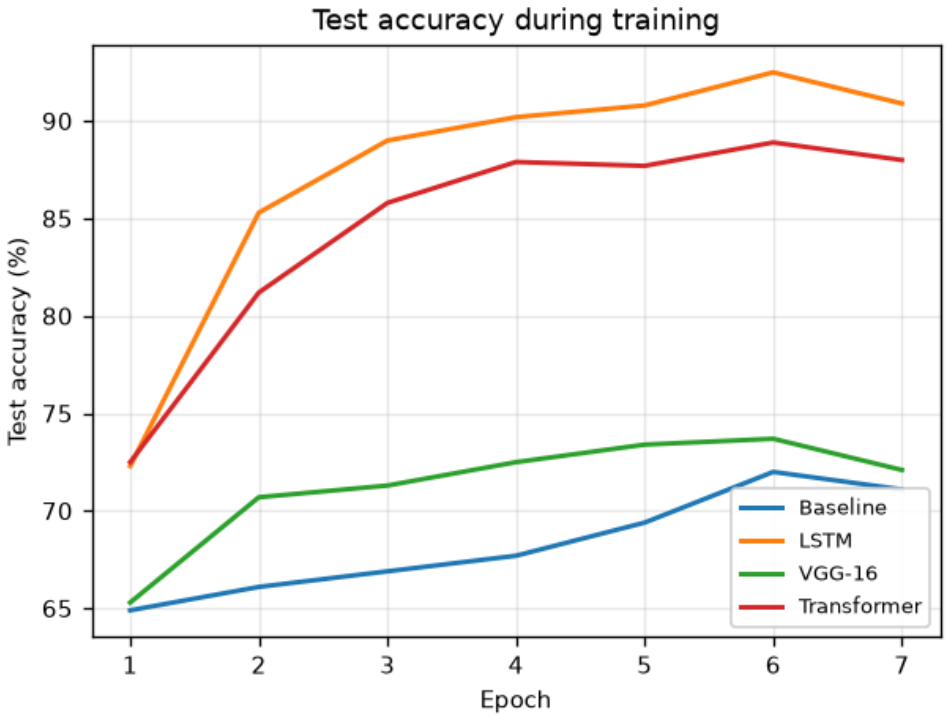


Figure 3: Test accuracy during training for the models in this experiment.

VGG-16 reaches 69.1% test accuracy at epoch 7 (Figure 3). Transformer outperforms Baseline by 3.1 percentage points at epoch 5. As shown in Figure 3, at epoch 4, VGG-16 achieves about 70% test accuracy.

Figure 4 shows the validation accuracy by dataset.

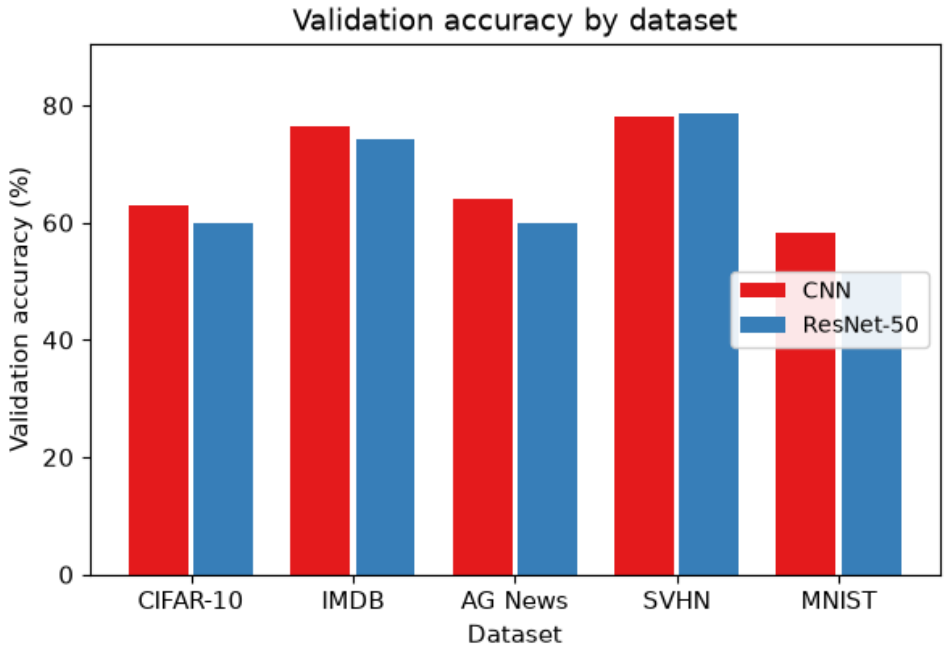


Figure 4: Validation accuracy by dataset for the models in this experiment.

As shown in Figure 4, on SVHN, ResNet-50 obtains a validation accuracy of over 75%. ResNet-50 achieves 74.3% validation accuracy on IMDB.

3 Discussion

On Fashion-MNIST, XGBoost obtains an accuracy of 56.4%. Overall, the results are in line with what we expected from the literature.